A A short introduction to analytic number theory

Section A.1 is mainly inspired from Jean Pierre Serre *A Course in Arithmetic* [Ser12] and Bruno Kahn's Nano textbook [Kah16, Chapter 4]. Only very basic and necessary materials are presented. We follow [Brü13] to prove a first version of the Prime Number Theorem in Section A.2. Most if not all of the proofs are to be found in the references. A notable exception is the proof of the Dirichlet theorem: a complete treatment is made in reason of the elegance of the proof. The presentation of the proof of the Dirichlet theorem in Section A.3 comes from a lecture of Anders Södergren in Chalmers (2025).

If not stated otherwise, p is a prime number.

A.1 Zeta and L-functions

It is well known that the Riemann zeta function ζ_{Riemann} encodes information about prime number and especially their repartition. A first avatar of this idea is to be found in the Euler product of ζ_{Riemann}

$$\zeta_{\text{Riemann}}(s) := \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} (1 - 1/p^s)^{-1}$$

as a consequence of the fundamental theorem of arithmetic. From the theory of generating functions, the Riemann zeta function can be seen as $\sum_{n\geq 1} a_n n^{-s}$ with $(a_n)_n$ the constant sequence such that each element is 1. It is then tempting to ask if and how properties of the Riemann zeta function are conserved for various over sequences (a_n). We study the case of Dirichlet characters and briefly present more general settings.

Dirichlet characters Let m be an integer.

Definition A.1. A Dirichlet character modulo m is a function $\chi : \mathbf{Z} \to \mathbf{C}$ such that it is:

(i) nonzero: $\chi \neq 0$,

(ii) periodic: $\chi(x + m) = \chi(x)$ for all x in $\mathbf{Z}$,

(iii) completely multiplicative: $\chi(xy) = \chi(x)\chi(y)$ for all x, y in $\mathbf{Z}$ and

(iv) it respects coprimes: $\chi(x) = 0$ if $(x, m) \neq 1$.

Note that Dirichlet characters are characters of $(\mathbf{Z}/m\mathbf{Z})^\times$ lifted up to $\mathbf{Z}$. We denote by $\chi_0(n)$ the trivial character (also called principal character) modulo m. It is defined such that

$$\chi_0(n) = \begin{cases} 1 & \text{if (n, q) = 1,} \\ 0 & \text{otherwise.} \end{cases}$$

With respect to pointwise multiplication, the set of Dirichlet characters obviously forms a group.

Dirichlet series Let $(\lambda_n)_n$ a increasing sequence of positive real numbers diverging.

Definition A.2. Let $a_n \in \mathbf{C}$ and $s \in \mathbf{C}$. A Dirichlet series is a series of the form

$$\sum_{n=1}^{\infty} a_n e^{-\lambda_n s}.$$

When $a_n = 1$ and $\lambda_n = \log(n)$ one finds the Riemann zeta function. This more general context is very fruitful and most of the classical results of convergence of complex series still hold true. More generally, before considering the case $a_n = \chi(n)$ for χ a Dirichlet character, we examine the case $a_n = f(n)$ for f a multiplicative bounded function.

Proposition A.3. *The Dirichlet series $\sum_{n=1}^{\infty} f(n)/n^s$ converges absolutely for $\mathrm{Re}(s) > 1$. Its sum on this domain is*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \text{ prime}} (1 + f(p)p^{-s} + \cdots + f(p^m)p^{-ms} + \ldots).$$

Moreover, if f is completely multiplicative (for instance, a Dirichlet character) we have

$$\sum_{n=1}^{\infty} f(n)/n^s = \prod_{p \text{ prime}} \frac{1}{1 - f(p)/p^s}.$$

L functions Let m be an integer and χ A Dirichlet character modulo m.

Definition A.4. The L function attached to a Dirichlet series is

$$\mathrm{L}(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

These functions are crucial in number theory and especially to prove Dirichlet theorem on primes in arithmetic progressions, especially see Theorem A.11 of Section A.3. They also are related to the Riemann zeta function in a natural way.

Proposition A.5. *Let χ_0 be the trivial character, then*

$$\mathrm{L}(s, \chi_0) = \zeta_{\text{Riemann}}(s) \prod_{p|m} (1 - p^{-s}).$$

More generally, let f be the conductor of χ a Dirichlet character modulo m, let S be the set of all prime factor of m not dividing f and let $\tilde{\chi}$ the primitive character modulo f associated, then

$$\mathrm{L}(s, \chi) = \mathrm{L}(s, \tilde{\chi}) \prod_{p \in S} \left(1 - \frac{\tilde{\chi}(p)}{p^s}\right).$$

This kind of result can be extended to various number fields, for $K = \mathbf{Q}(\mu_n)$ see for instance [Kah16, Section 1.4]. To get the Euler product, Proposition A.3 can be generalised the following way.

Proposition A.6. *Let χ be a non principal character, the series $\mathrm{L}(s, \chi)$ converges (resp. converges absolutely) in the half-plane $\mathrm{Re}(s) > 0$ (resp. $\mathrm{Re}(s) > 1$). We have*

$$\mathrm{L}(s, \chi) = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p)p^{-s}}, \quad \mathrm{Re}(s) > 1.$$

We conclude by one of the main properties of L functions, their functional equation. We follow [Kum11, Chapter 1], see also the more general [Mic22, Section 2]. Denote $\Lambda(s, \chi) = \mathrm{L}(s, \chi)\mathrm{L}_\infty(s, \chi)$ the completed L function, see Tate thesis for an abstract explanation of the place at infinity factor. The following functional equation is satisfied

$$\Lambda(s, \chi) = \kappa(\chi)\Lambda(1 - s, \overline{\chi}) \tag{63}$$

with $\kappa(\cdot)$ a complex number with modulus one. The classical example is the functional equation of the Riemann zeta function. From the Poisson summation formula and the theory of theta functions, one finds $\Lambda(s, \mathbf{1}) = \Lambda(1 - s, \mathbf{1})$ with $\Lambda(s, \mathbf{1}) = \pi^{-s/2}\Gamma(s/2)\zeta_{\text{Riemann}}(s)$.

A.2 Prime number theorem

We first prove a weak version of the prime number theorem [Brü13, Section 1.7] assuming some facts about Chebyshev's function and then state the classical prime number theorem with error term [Brü13, Section 2.8].

Theorem A.7. *There exists a constant $C > 0$ such that we have*

$$\pi(x) = \int_2^x \frac{\mathrm{d}u}{\log u} + \mathcal{O}\Big(x \exp\Big(-C(\log x)^{1/10}\Big)\Big).$$

Proof. Recall that the Chebyshev ψ function can be expressed using the von Mangoldt Λ function: $\psi(x) = \sum_{n \leq x} \Lambda(n)$. Let $c > 1$, $T \geq 2$ and $x \geq 2$. From Perron's formula and contour integration, we have

$$\psi(x) = \frac{-1}{2i\pi} \int_{c-iT}^{c+iT} \frac{\zeta'(s)}{\zeta(s)} x^s \frac{\mathrm{d}s}{s} + \mathcal{O}\left(\frac{x^c}{T}\left|\frac{\zeta'(c)}{\zeta(c)}\right| + \log x + \frac{x(\log x)^2}{T}\right) \tag{64}$$

with

$$-\int_{c-iT}^{c+iT} \frac{\zeta'(s)}{\zeta(s)} x^s \frac{\mathrm{d}s}{s} = 2i\pi x + \left(\int_{\gamma_1} + \int_{\gamma_2} + \int_{\gamma_3}\right) \frac{\zeta'(s)}{\zeta(s)} x^s \frac{\mathrm{d}s}{s} \tag{65}$$

where

$$\gamma_1 = [c + iT, c' + iT], \quad \gamma_2 = [c' + iT, c' - iT], \quad \gamma_3 = [c' - iT, c - iT]$$

for a convenient c' depending on a parameter λ, see [Brü13, p. 40]. Combining (64) and (65), one finds

$$\psi(x) = x + \mathcal{O}\left(\frac{x^{1+\lambda}}{T\lambda} + \log x + \frac{x(\log x)^2}{T} + x^{1-\lambda}(\log T)^{10}\right).$$

Let $T = \exp(\delta^{-1/9}(\log x)^{1/10})$, one then has

$$\psi(x) = x + \mathcal{O}\Big(x \exp\Big(-\delta^3(\log x)^{1/10}\Big)\Big).$$

By the definition of ψ and θ functions, we get

$$\theta(u) = \sum_{p \leq u} \log p = u + \mathcal{O}\Big(u \exp\Big(-\delta^3(\log u)^{1/10}\Big)\Big).$$

We then conclude by a summation by parts

$$\begin{aligned}
\pi(x) = \sum_{u=2}^{x} \mathbb{1}_{u \text{ prime}} \log u \frac{1}{\log u} &= \frac{\theta(x)}{\log x} - \int_2^x \theta(u) \frac{\mathrm{d}}{\mathrm{d}u}\left(\frac{1}{\log u}\right) \mathrm{d}u \\
&= \frac{x}{\log x} - \int_2^x u \frac{\mathrm{d}}{\mathrm{d}u}\left(\frac{1}{\log u}\right) \mathrm{d}u + \mathcal{O}\Big(x \exp\Big(-\delta^3(\log x)^{1/10}\Big)\Big) \\
&= \frac{x}{\log x} + \int_2^x \frac{1}{(\log u)^2} \mathrm{d}u + \mathcal{O}\Big(x \exp\Big(-\delta^3(\log x)^{1/10}\Big)\Big).
\end{aligned}$$

and by an integration by parts we get

$$\int_2^x \frac{1}{(\log u)^2} \mathrm{d}u = \mathrm{li}(x) - \mathrm{li}(2) - \frac{x}{\log x} + \frac{2}{\log 2}.$$

This concludes the proof. □

The exponent $1/10$ can be improved to $1/2$.

Theorem A.8 (Prime number theorem with error term)**.** *There exists a constant $C > 0$ such that we have*

$$\pi(x) = \int_2^x \frac{\mathrm{d}u}{\log u} + \mathcal{O}\Big(x \exp\Big(-C\sqrt{\log x}\Big)\Big).$$

A.3 Primes in arithmetic progressions

In order to prove the Dirichlet theorem (Theorem A.9), we study the behaviour of an arithmetic-interest function (Theorem A.10).

Theorem A.9. *If $a \in \mathbf{Z}$, $q \in \mathbf{N}$ are such that $(a, q) = 1$, then $\sum_{p\equiv a \bmod q} \frac{1}{p} = +\infty$, i.e. there are infinitely many primes in arithmetic progression subject to the congruence condition.*

Theorem A.10. *Under the same assumptions as in Theorem A.9, we have*

$$\sum_{\substack{p\le x\\ p\equiv a \bmod q}} \frac{1}{p} = \frac{1}{\phi(q)} \log\log x + \mathcal{O}_q(1) \quad \text{for } x \geq 3.$$

Theorem A.11. *Let $q \in \mathrm{N}$, if $\chi \neq \chi_0$ is a Dirichlet character modulo q, then $\mathrm{L}(1, \chi) \neq 0$.*

We present a complete proof of Theorem A.10, the classical Dirichlet theorem on primes in arithmetic progressions (Theorem A.9) is an immediate corollary. Theorem A.11 is crucial to prove Theorem A.10 and will be assumed at first and proved afterwards.

Proof of Theorem A.10. Let $a \in \mathbf{Z}$ and $q \in \mathbf{N}$ be such that $(a, q) = 1$. Define the following quantities :

$$S_{a,q}(x) = \sum_{\substack{p\le x\\ p\equiv a \bmod q}} \frac{1}{p} \quad \text{for } x \geq 2, \quad F_{a,q}(s) = \sum_{p\equiv a \bmod q} \frac{1}{p^s} \quad \text{for } \mathrm{Re}(s) > 1.$$

Recall that in the upper half-plane $\mathrm{Re}(s) > 1$, there is absolute and uniform convergence on compact subsets. We will have to go through six steps.

Step 1 It is sufficient to show that

$$F_{a,q}(\sigma) = \frac{1}{\phi(q)} \log \frac{1}{\sigma - 1} + \mathcal{O}_q(1), \quad 1 < \sigma \leq 2. \tag{66}$$

Later on, we will see that the bound 2 is essentially decorative. The major problems arise near $s = 1$. Let $x \geq 3$ and choose $\sigma := \sigma(x) = 1 + \frac{1}{\log x}$. With such σ in (66), we get the same main terms. It remains to show that

$$\left|S_{a,q}(x) - F_{a,q}(\sigma(x))\right| = \mathcal{O}_q(1).$$

Note that

$$S_{a,q}(x) - F_{a,q}(\sigma(x)) = \sum_{\substack{p\le x\\ p\equiv a \bmod q}} \frac{1 - p^{1-\sigma(x)}}{p} - \sum_{\substack{p>x\\ p\equiv a \bmod q}} \frac{1}{p^{\sigma(x)}}.$$

Moreover $1 - p^{1-\sigma(x)} = 1 - e^{(1-1-1/\log x)\log p} = 1 - e^{-\log p/\log x} \ll \log p/\log x$ by a Taylor expansion for $p \leq x$. Hence

$$\sum_{\substack{p\le x\\ p\equiv a \bmod q}} \frac{1 - p^{1-\sigma(x)}}{p} \ll \frac{1}{\log x} \sum_{p\le x} \frac{\log p}{p} \ll 1$$

by Mertens theorem. We now need summation by parts (always in the bag pocket) to control the last part (the sum for $p > x$). Thus

$$\sum_{\substack{p>x\\ p\equiv a \bmod q}} \frac{1}{p^{\sigma(x)}} \leq \sum_{p>x} \frac{1}{p^{\sigma(x)}} = -\frac{\pi(x)}{x^{\sigma(x)}} + \sigma(x) \int_x^\infty \frac{\pi(u)}{u^{\sigma(x)+1}} \mathrm{d}u \ll \frac{1}{\log x} + \int_x^\infty \frac{1}{u^{\sigma(x)+1} \log u} \mathrm{d}u$$

$$\ll \frac{1}{\log x} + \frac{1}{\log x} + \int_x^\infty \frac{1}{u^{\sigma(x)+1}} \mathrm{d}u \ll \frac{1}{\log x} + \frac{1}{\log x} \frac{x^{1-\sigma(x)}}{\sigma(x) - 1} \ll 1$$

for x sufficiently large. Notice that, in the second line, we used Chebyshev theorem or the Prime Number Theorem; and, in the third line, we used that $1/\log x$ is non-increasing ($x \geq 3$). Thus

$$\left|S_{a,q}(x) - F_{a,q}(\sigma(x))\right| = \mathbf{O}_q(1)$$

and this shows this is sufficient to study the behaviour of $F_{a,q}(\sigma)$.

Step 2 Let χ be a Dirichlet character modulo q, define

$$F_\chi(s) = \sum_p \frac{\chi(p)}{p^s}, \quad \mathrm{Re}(s) > 1.$$

Never forget that, in this domain, we have absolute and uniform convergence of the series. Letting $\sigma > 1$, one has

$$F_{a,q}(\sigma) = \sum_{p \equiv a \bmod q} \frac{1}{p^\sigma} = \sum_p \frac{1}{p^\sigma} \frac{1}{\phi(q)} \sum_{\chi \bmod q} \chi(p)\overline{\chi(a)} = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} \sum_p \frac{\chi(p)}{p^\sigma} = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} F_\chi(\sigma).$$

We applied a very prolific strategy to make congruences disappear : use orthogonality relations of Dirichlet series! And, again, we use the fact that there is only a finite number of Dirichlet characters modulo q (there are exactly $\phi(q)$) and that we can rearrange a finite sum of absolutely convergent series. Thus, one do not care about summation order.

Step 3 We need to express $F_\chi(s)$ in terms of $\log \mathrm{L}(s,\chi)$. Recall the logarithmic formula

$$\log \mathrm{L}(s,\chi) = -\sum_p \log\left(1 - \frac{\chi(p)}{p^s}\right).$$

It is a direct consequence of the Euler product formula for such L functions. We now use Taylor expansion

$$\log \mathrm{L}(s,\chi) = \sum_p \sum_{m=1}^{\infty} \frac{\chi(p)^m}{mp^{ms}} = \sum_p \frac{\chi(p)}{p^s} + \sum_p \sum_{m=2}^{\infty} \frac{\chi(p)^m}{mp^{ms}} = F_\chi(s) + \text{remainder}.$$

We were able to make such an expansion because $|\chi(p)/p^s| \leq |1/p^s| \leq 1/2 < 1$. Estimate the remainder trivially

$$\left|\sum_p \sum_{m=2}^{\infty} \frac{\chi(p)^m}{mp^{ms}}\right| \leq \sum_p \sum_{m=2}^{\infty} \frac{1}{mp^m} \leq \frac{1}{2} \sum_p \frac{1}{p^2} \sum_{m=0}^{\infty} \frac{1}{p^m} \leq \frac{1}{2} \sum_{n \geq 2} \frac{1}{n^2} \sum_{m=0}^{\infty} \frac{1}{n^m} \frac{1}{2} \sum_{n \geq 2} \frac{1}{n^2} \frac{n}{n-1} \leq \frac{1}{2} \sum_{n \geq 2} \frac{1}{n(n-1)} = \mathcal{O}(1).$$

We conclude that

$$F_\chi(s) = \log \mathrm{L}(s,\chi) + \mathcal{O}(1), \quad \sigma > 1.$$

Step 4 We now prove the following statement

$$\exists \sigma_0 > 1, \ \log \mathrm{L}(s,\chi_0) = \log \frac{1}{\sigma - 1} + \mathcal{O}_q(1), \quad 1 < \sigma \leq \sigma_0.$$

Note that σ_0 depends on q. If one wonders why proving such statement, think about step 1! Recall that $L(s,\chi_0)$ is analytic in $\sigma > 0$ apart from the simple pole $s = 1$, with residue $\phi(q)/q$. Thus

$$\mathrm{L}(s,\chi_0) = \frac{\phi(q)}{q} \frac{1}{s-1} + A(s)$$

where A is analytic in $\sigma > 0$. Since, $A(s)$ is bounded on compacts in $\sigma > 0$, we get

$$|A(\sigma)| \le \frac{\phi(q)}{2q(\sigma-1)}, \quad 1 < \sigma \le \sigma_0 \tag{67}$$

for some $\sigma_0 := \sigma_0(q) > 1$. We decorated the bound such that the incoming computations look nice. Pay attention that $]1, \sigma_0]$ is not compact! It has no repercussion there since, if σ is close to 1, the quantity $1/(\sigma-1)$ will be large. Thus, using log properties and tricking quantities

$$\begin{aligned}\log \mathrm{L}(s,\chi) &= \log\left(\frac{\phi(q)}{q}\frac{1}{\sigma-1}\left(1+A(\sigma)(\sigma-1)\frac{q}{\phi(q)}\right)\right) = \log\left(\frac{\phi(q)}{q}\frac{1}{\sigma-1}\right) + \log\left(1+A(\sigma)(\sigma-1)\frac{q}{\phi(q)}\right)\\ &= \log\left(\frac{\phi(q)}{q}\right) + \log\left(\frac{1}{\sigma-1}\right) + \log\left(1+A(\sigma)(\sigma-1)\frac{q}{\phi(q)}\right).\end{aligned}$$

We will restrict to real numbers, otherwise we should care about branches of the logarithm. Note that, due to (67), we have

$$\left|A(\sigma)(\sigma-1)\frac{q}{\phi(q)}\right| \le \frac{1}{2}.$$

Hence

$$\log \mathrm{L}(s,\chi_0) = \log\frac{1}{\sigma-1} + \mathcal{O}_q(1), \quad 1 < \sigma \le \sigma_0.$$

From step 3, we deduce

$$F_{\chi_0}(\sigma) = \log \mathrm{L}(s,\chi_0) + \mathcal{O}(1) = \log\frac{1}{\sigma-1} + \mathcal{O}(1), \quad 1 < \sigma \le \sigma_0. \tag{68}$$

Also, (68) holds more generally for any $\sigma > 1$, if we remove the log term for large σ

$$\left|F_{\chi_0}(\sigma)\right| = \left|\sum_p \frac{\chi_0(p)}{p^\sigma}\right| \le \sum_p \frac{1}{p^\sigma} = \mathcal{O}_q(1), \quad \sigma > \sigma_0. \tag{69}$$

Step 5 We now prove the following statement on the remaining characters : if $\chi \ne \chi_0$ is a Dirichlet character modulo q, then $F_\chi(\sigma) = \mathcal{O}_q(1)$ in $\sigma > 1$. Recall that, if $\chi \ne \chi_0$, then $\mathrm{L}(s,\chi)$ is analytic in $\sigma > 0$ (the situation is even better than the χ_0 case). Moreover, from Theorem A.11, we have $\mathrm{L}(s,\chi) \ne 0$ for $\chi \ne \chi_0$. Thus, $\log \mathrm{L}(s,\chi)$ is well-defined, analytic and then continuous in a neighbourhood of $s = 1$. Hence, there exists a $\sigma_0 > 1$ such that $\log \mathrm{L}(s,\chi)$ is bounded in $1 < \sigma \le \sigma_0$. From step 3, we deduce

$$F_\chi(s) = \mathcal{O}_q(1), \quad 1 < \sigma \le \sigma_0$$

The domain can be extended to $\sigma > 1$ by a same argument as in (69).

Step 6 We now put the pieces together. From step 2, we have

$$F_{a,q}(\sigma) = \sum_{p \equiv a \bmod q} \frac{1}{p^\sigma} = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \overline{\chi(a)} F_\chi(\sigma) = \frac{1}{\phi(q)} \overline{\chi_0(a)} F_{\chi_0}(\sigma) + \mathcal{O}_q(1) = \frac{1}{\phi(q)} \log\frac{1}{\sigma-1} + \mathcal{O}_q(1)$$

where the $\mathcal{O}_q(1)$ is the contribution from non trivial characters (step 5). Theorem A.10 is then proved. □

We now prove Theorem A.11 about the non-vanishing of Dirichlet L functions at $s = 1$ for $\chi \ne \chi_0$.

Proof of Theorem A.11. We consider two cases : complex and real characters.

Complex characters Recall that a character is complex if there exists an $n \in \mathbf{Z}$ such that $\chi(n) \notin \mathbf{R}$. Note that for $\sigma > 1$, we have

$$\sum_{\chi \bmod q} \log \mathrm{L}(s,\chi) = -\sum_{\chi \bmod q}\sum_p \log\left(1-\frac{\chi(p)}{p^s}\right) = \sum_{\chi \bmod q}\sum_p\sum_{m=1}^{\infty}\frac{\chi(p)^m}{mp^{ms}} = \sum_p\sum_{m=1}^{\infty}\frac{1}{p^{ms}}\sum_{\chi \bmod q}\chi(p)^m.$$

However, by the orthogonality relations, we have

$$\sum_{\chi \bmod q}\chi(p)^m = \sum_{\chi \bmod q}\chi(p^m) = \begin{cases}\phi(q) & \text{if } p^m \equiv 1 \bmod q\\ 0 & \text{else.}\end{cases}$$

Thus, since we sum non-negative terms, we conclude

$$\sum_{\chi \bmod q}\log \mathrm{L}(\sigma,\chi) \geq 0, \quad \sigma > 1.$$

We have already done these calculations in Step 3. Take the exponential to get

$$\prod_{\chi \bmod q}\mathrm{L}(\sigma,\chi) \geq 1, \quad \sigma > 1. \tag{70}$$

Now be really careful (*ne pas aller trop vite en besogne!*), suppose $\mathrm{L}(1,\chi_1) = 0$ for a complex character χ_1. Then

$$L(1,\overline{\chi_1}) = \sum_{n=1}^{\infty}\frac{\overline{\chi_1}(n)}{n} = \overline{\sum_{n=1}^{\infty}\frac{\chi_1(n)}{n}} = \overline{\mathrm{L}(1,\chi_1)}$$

by absolute convergence. Note that χ_1 and $\overline{\chi_1}$ are distinct since χ_1 is a complex character. We thus have at least two factors in (70) that are zero at $s = 1$. Next, write

$$\prod_{\chi \bmod q}\mathrm{L}(s,\chi) = \mathrm{L}(s,\chi_0)\mathrm{L}(s,\chi_1)\mathrm{L}(s,\overline{\chi_1})\prod_{\chi \bmod q \chi\neq\chi_0,\chi_1,\overline{\chi_1}}\mathrm{L}(s,\chi). \tag{71}$$

We have four different terms on the right hand side : a **simple** pole at $s = 1$, a **double** zero at $s = 1$ and an analytic part at $s = 1$. The whole product is meromorphic in $\sigma > 0$ and (71) implies that $s = 1$ is a removable singularity. Then, the product will extend to an analytic function in $\sigma > 0$ with a zero at $s = 1$. Taking the limit, we get

$$\lim_{s\to 1^+}\prod_{\chi \bmod q}\mathrm{L}(s,\chi) = 0.$$

This contradicts (70). Then, the assertion is proved for complex characters.

Real characters It is a little bit harder. Again, suppose $\mathrm{L}(s,\chi) = 0$. Remark that $\mathrm{L}(s,\chi)\mathrm{L}(s,\chi_0)$ is analytic at $s = 1$ so in $\sigma > 0$ (its only pole has been cancelled). Also, $\mathrm{L}(2s,\chi_0)$ is analytic in $\sigma > 1/2$ and non-zero. Introduce now the analytic function $\Psi(s) = \mathrm{L}(s,\chi)\mathrm{L}(s,\chi_0)/\mathrm{L}(2s,\chi_0)$ for $\sigma > 1/2$. Note that $\lim_{s\to 1/2^+}\mathrm{L}(2s,\chi_0) = +\infty$, thus $\lim_{s\to 1/2^+}\Psi(s) = 0$ because the denominator is finite when $s \to 1/2^+$. To find a contradiction, we need to understand Ψ better. For $\sigma > 1$, we use the Euler products

$$\Psi(s) = \prod_p \frac{\left(1-\frac{\chi(p)}{p^s}\right)^{-1}\left(1-\frac{\chi_0(p)}{p^s}\right)^{-1}}{\left(1-\frac{\chi_0(p)}{p^{2s}}\right)^{-1}} = \prod_p \lambda(\chi,p,s).$$

Note that $\chi_0(p) = 1$ if $p \nmid q$, $\chi_0(p) = 0$ if $p|q$ and $\chi(p) = \pm 1$ if $p \nmid q$, $\chi(p) = 0$ if $p|q$ because χ is a real character. Thus

$$\lambda(\chi, p, s) = \begin{cases} 1 & \text{if } p|q \\ \frac{\left(1+\frac{\chi(p)}{p^s}\right)^{-1}\left(1-\frac{\chi_0(p)}{p^s}\right)^{-1}}{\left(1-\frac{\chi_0(p)}{p^{2s}}\right)^{-1}} = 1 & \text{if } p \nmid q \text{ and } \chi(p) = -1 \\ \frac{\left(1-\frac{\chi(p)}{p^s}\right)^{-1}\left(1-\frac{\chi_0(p)}{p^s}\right)^{-1}}{\left(1-\frac{\chi_0(p)}{p^{2s}}\right)^{-1}} & \text{else.} \end{cases}$$

As a direct consequence, for $\sigma > 1$, we have

$$\Psi(s) = \prod_{p,\ \chi(p)=1} \frac{\left(1 - \frac{\chi(p)}{p^s}\right)^{-1}\left(1 - \frac{\chi_0(p)}{p^s}\right)^{-1}}{\left(1 - \frac{\chi_0(p)}{p^{2s}}\right)^{-1}} = \prod_{p,\ \chi(p)=1} \left(\frac{1 + \frac{1}{p^s}}{1 - \frac{1}{p^s}}\right) = \prod_{p,\ \chi(p)=1} \left(1 + \frac{1}{p^s}\right)\left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \dots\right)$$

using the geometric series expansion. Note that such prime ($\chi(p) = 1$) exists, otherwise the product would be empty and then equal to 1. Taking the limit, it would still be 1. Moreover, we would be able to extend the product to a large region of the complex problem (no problem since it would be a constant function). This is a contradiction since the limit of $\Psi(s)$ is 0 as long as $s \to 1/2^+$ and not 1. Now, we claim and give only a rough but nearly complete idea of the proof that $\Psi(s)$ can be expanded as a Dirichlet series $\Psi(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ for $\sigma > 1$ where $a_1 = 1$ and $a_n \geq 0$ for all $n \in \mathbf{N}$. This series is also convergent in $\sigma > 1$ and uniformly convergent on compact subsets of $\sigma > 1$. It is not trivial since we start from a product and go to a sum (usually, we do the contrary). When $\sigma > 1$, we have $|1/p^s| < 1$ for all p and we compute

$$\Psi(s) = \prod_{p,\ \chi(p)=1} \left(1 + 2p^{-s} + 2p^{-2s} + 2p^{-3s} + \dots\right).$$

Define, for fixed $s \in \mathbf{C}$ with $\sigma > 1$, $f : \mathbf{N} \to \mathbf{C}$ to be multiplicative and satisfying

$$f(p^k) = \begin{cases} 0 & \text{if } \chi(p) = 0 \\ 2p^{-ks} & \text{if } \chi(p) = 1. \end{cases}$$

Then

$$f(n) = n^{-s} \prod_{p|n} \begin{Bmatrix} 0 & \text{if } \chi(p) = 0 \\ 2p^{-ks} & \text{if } \chi(p) = 1. \end{Bmatrix}$$

One still has to prove that the Dirichlet series defined is absolutely convergent. We conclude that

$$\Psi(s) = \sum_{n=1}^{\infty} f(n) = \sum_{n=1}^{\infty} n^{-s} \prod_{p|n} \begin{Bmatrix} 0 & \text{if } \chi(p) = 0 \\ 2p^{-ks} & \text{if } \chi(p) = 1. \end{Bmatrix}$$

and verify that $f(1) = 1$ since the product is empty and $f(n) \geq 0$. Now, recall that Ψ is analytic in the half-plane $\sigma > 1/2$. Hence, because of analyticity, we can expand Ψ in power series around $s = 2$ (say, somewhere in $\sigma > 1/2$) with radius of convergence at most[1] $3/2$

$$\Psi(s) = \sum_{m=0}^{\infty} \frac{1}{m!} \Psi^{(m)}(2)(s-2)^m.$$

We need to understand the derivatives of Ψ. Note that we can differentiate termwise since it is a Dirichlet series convergent in $\sigma > 1$. We have

$$\Psi^{(m)}(2) = (-1)^m \sum_{n=1}^{\infty} a_n (\log n)^m n^{-2} = (-1)^m b_m$$

[1]Verify! I guess this is at most but I wrote at least.

where b_m is non-negative. Thus

$$\Psi(s) = \sum_{m=0}^{\infty} \frac{1}{m!} b_m (2-s)^m.$$

Notice that the $(-1)^m$ was put inside $(s-2)^m$. Hence, if we restrict to real s such that $1/2 < s < 2$, then all terms in the Taylor series are non-negative and thus $\Psi(s) \geq \Psi(2) \geq 1$ by the product formulas. In such case, Ψ could never tend to 0. This is a contradiction (of the $1/2^+$ limit). As a conclusion, if χ is a real or complex character different from χ_0, then $\mathrm{L}(1,\chi) \neq 0$. □

A. Analytic continuation and functional equation. Let χ be a Dirichlet character modulo q. We need to express $n \mapsto \chi(n)$ as a linear combination of $n \mapsto e^{2i\pi mn/q}$ for $m = 0, \ldots, q-1$.

For any χ modulo q, we define the Gauss sum $\tau(\chi)$ by

$$\tau(\chi) = \sum_{m \in \mathrm{Z}/q\mathrm{Z}} \chi(m) e^{2i\pi m/q}.$$

If $(n,q) = 1$, then

$$\chi(n)\tau(\overline{\chi}) = \chi(n) \sum_{m \in \mathrm{Z}/q\mathrm{Z}} \overline{\chi}(m) e^{2i\pi m/q} = \sum_{m \in \mathrm{Z}/q\mathrm{Z}} \overline{\chi}(n^{-1}m) e^{2i\pi m/q} = \sum_{h \in \mathrm{Z}/q\mathrm{Z}} \overline{\chi}(h) e^{2i\pi hn/q}.$$

This is the desired decomposition of $\chi(n)$.

Note that, if χ is a primitive character, then $|\tau(\chi)| = \sqrt{q}$, more generally see Theorem 1.1.

Theorem A.12. *Let χ be a primitive character modulo $q \geq 3$. Then $\mathrm{L}(s,\chi)$ has an analytic continuation to an entire function. Furthermore, $\mathrm{L}(s,\chi)$ satisfies*

$$\xi(1-s,\overline{\chi}) = \frac{i^a \sqrt{q}}{\tau(x)} \xi(s,\chi)$$

where $\xi(s,\chi) = \left(\frac{\pi}{a}\right)^{-1/2(s+a)} \Gamma(1/2(s+a)) \mathrm{L}(s,\chi)$, $a = 0$ if $\chi(-1) = 1$ and $a = 1$ if $\chi(-1) = -1$. Note that the absolute value of the factor in the functional equation is 1.

The proof is very similar to the one for $\zeta(s)$. In particular, we start looking at the function $(s,\chi) \mapsto \pi^{-s/2} q^{s/2} \Gamma(s/2) L(s,\chi)$ and representing it with an integral (the Γ integral representation). Then, it is exactly the same idea but not the same details (the theta identity is a little bit trickier).

Remark that we get information about zeros of $\xi(s,\chi)$ for $0 \leq \sigma \leq 1$ and $\mathrm{L}(s,\chi)$ (trivial and non trivial).

B. Zero-free regions for $\mathrm{L}(s,\chi)$**.** It is relatively easy to generalize what is done for $\zeta(s)$ to $\mathrm{L}(s,\chi)$ with a **fixed** χ modulo q. However, we would like estimates with explicit q-dependence. This is way more difficult.

Theorem A.13. *i) There exists an absolute constant $c > 0$ such that, for all q in* N *and every* ***complex*** *character modulo q, $\mathrm{L}(s,\chi)$ has no zeros in the following region :*

$$\sigma \geq \begin{cases} 1 - \frac{c}{\log(q|t|)} & |t| \geq 1, \\ 1 - \frac{c}{\log q} & |t| \leq 1. \end{cases}$$

ii) There exists an absolute constant $c > 0$ such that, for all q in N *and every* ***real non trivial*** *character modulo q, $\mathrm{L}(s,\chi)$ has at most one zero in the same σ region. If such a zero exist, then this is a simple real zero (and it is a potential threat to any generalization of the Riemann hypothesis). These zeros are called Siegel zeros.*

C. The PNT for arithmetic progressions. This is an asymptotic formula for

$$\pi(x; q, a) = \#\{p \leq x \ : \ p \equiv a \bmod q\}$$

(p is obviously a prime). As for the PNT, it is more convenient to work with a Ψ function. We introduce

$$\Psi(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a \bmod q}} \Lambda(n)$$

where Λ is the von Mongoldt function. How to relate Ψ with characters ? Obviously, use orthogonality ! If $(a, q) = 1$, then

$$\Psi(x; q, a) = \frac{1}{\phi(q)} \sum_{\chi \bmod q} \overline{\chi}(a) \Psi(x, \chi)$$

with $\Psi(x, \chi) = \sum_{n \leq x} \chi(n) \Lambda(n)$ is the Ψ function related to $\mathrm{L}(s, \chi)$.

Theorem A.14. *There exists an absolute constant $c_1 > 0$ such that, for all $x \geq 2$ and all pairs $\{q, a\}$ with $(a, q) = 1$, we have*

$$\Psi(x; q, a) = \frac{x}{\phi(q)} - \frac{\overline{\chi_1}(a) x^{\beta_1}}{\phi(q) \beta_1} + O(x e^{-c_1 \sqrt{\log x}})$$

where χ_1 is a real character modulo q (if it exists) for which $\mathrm{L}(s, \chi_1)$ has a Siegel zero β_1. If there is no Siegel zero, the second terms of the right hand side ***should*** *be removed.*

We know very few things about Siegel zeros.

Theorem A.15. *Let $c_1 > 0$ be as in Theorem A.14. For all $x \geq 2$ and all admissible a and q, we have*

$$\pi(x; q, a) = \frac{1}{\phi(q)} \mathrm{L}(x) - \frac{\overline{\chi}(a)}{\phi(q)} \mathrm{Li}(x^{\beta_1}) + O(x e^{-c_1 \sqrt{\log x}})$$

Cf. Montgomery's book for further results.